Best Practices for Biomedical Research Data Management

Module 2: Research Lifecycle
Data Types, Formats and Stages

Glenn Gaudette, PhD, Worcester Polytechnic Institute

This video will walk through the data types and formats that are generated over multiple stages of a biomedical engineering research project. The purpose of this research is to improve the methods to deliver cell therapy to improve cardiac function. The research is testing the delivery of cells via biological sutures implanted in rat hearts. The video illustrates the complexity of data types and file formats that can be involved in a single research project and the need to have a well thought-out strategy to manage all this data that can be clearly described.

BY NC SA

Pre-Surgery

Day = -2

Process: Incubate MSCs in qDot solution

Data Collected:

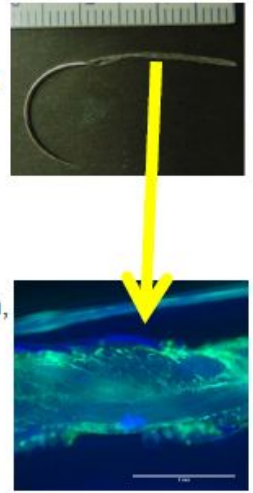
- Lot# for MSCs
- Passage # for MSCs
- Confirm qDot loading
  - Fluorescent images on inverted scope
    - Phase contrast image
    - Fluorescent image (red)
    - Overlap image (analyzed)
    - Other
      - » Objective used
      - » Scale bar on image
      - » Multiple images per well

Day = -1

Process: Load MSCs on biological suture

Data Collected:

- Concentration of cells in solution
- Volume delivered
- Confirm cell loaded on sutures
  - Fluorescent images on upright scope
    - Phase contrast image
    - Fluorescent images (green, blue, red)
    - Overlap image (analyzed)
    - Other
      - » Objective used
      - » Scale bar on image
      - » Multiple locations on each suture
      - » Antibodies used (lot#)



Surgery

Day = 0

Pre-operation

Day = 0

Intra-operation

Day = 0

Post-operation

Data Collected:

- Weight
- Drugs administered
  - Initial Heart rate; O_2 Sat
- Fix extra cell loaded biological suture in paraformaldehyde

Data Collected:

- Gross images of heart and suture placement
- Deliver cell loaded biological suture
- Drugs administered
- Heart rate; O_2 Sat

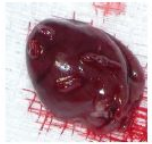
Data Collected:

- Heart rate; O_2 Sat
- Confirm cell loading on extra biological suture
  - Stain with Hoechst (nuclei) & Phalloidin (f-actin)
  - Fluorescent images on upright scope (same as Day = -1)

Post-Surgery

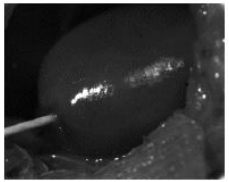
Day = 7

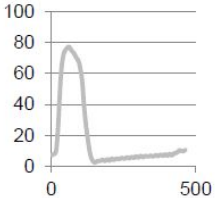
Process: Acquire Data for Future Processing



Data Collected:

- Gross images of heart
- Assessment of function
  - Left ventricular pressure
  - Oxygen saturation levels
  - High speed imaging of heart
- Images of heart after bisected



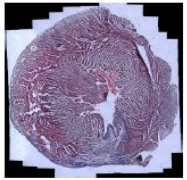


Day = 7+

Process: Histological Analysis

Data Analyzed:

- Heart section #
- Stained images (H&E or Masson's trichrome)
- Images on upright scope (see Day = -1)
- IHC stained images
  - Images on upright scope (see Day = -1)
- Confocal images



Day = 7+

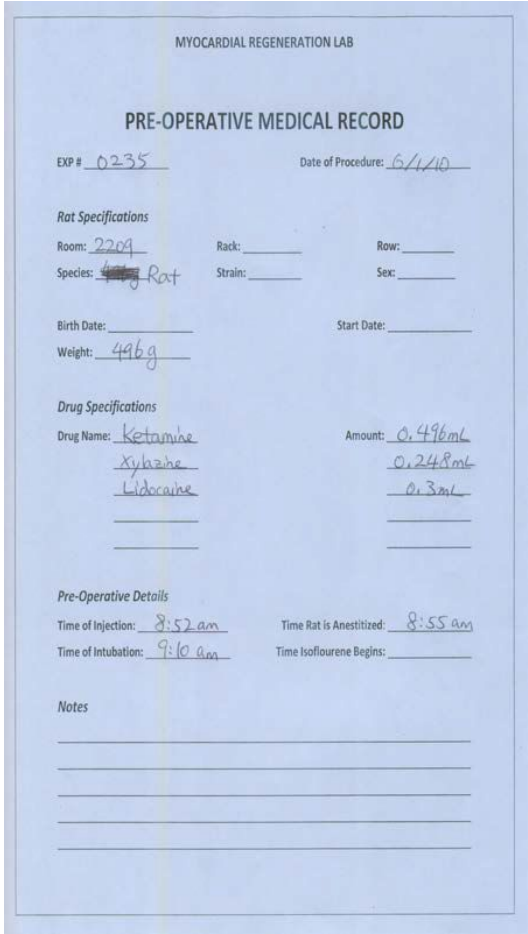
Process: Mechanical Function Analysis

Data Analyzed:

- Analyzed high speed images
  - Displacements between images
  - Displacements over entire heart cycle
  - Correlation of pressure with displacements
  - Displacements within small regions of the heart

Written Records

- Pre-operative record
  - Weight, heart rate, oxygen saturation
  - Drug dosage
- Lot# for MSCs
- Passage # for MSCs
- Time of procedure
  - Incisions
  - Cell delivery
- Volume delivered
- Drugs administered
- Heart rate; O2Sat



MYOCARDIAL REGENERATION LAB

PRE-OPERATIVE MEDICAL RECORD

EXP # 0235 Date of Procedure: 6/1/10

Rat Specifications

Room: 2209 Rack: _____ Row: _____

Species: [redacted] Rat Strain: _____ Sex: _____

Birth Date: _____ Start Date: _____

Weight: 496g

Drug Specifications

Drug Name:	Amount:
Ketamine	0.496mL
Xylazine	0.248mL
Lidocaine	0.3mL

Pre-Operative Details

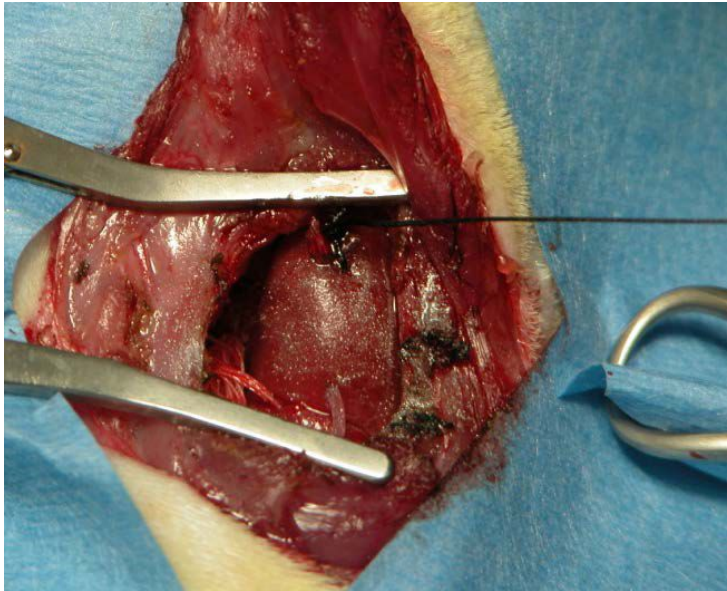
Time of Injection: 8:52am Time Rat is Anestitized: 8:55am

Time of Intubation: 9:10 am Time Isoflourene Begins: _____

Notes

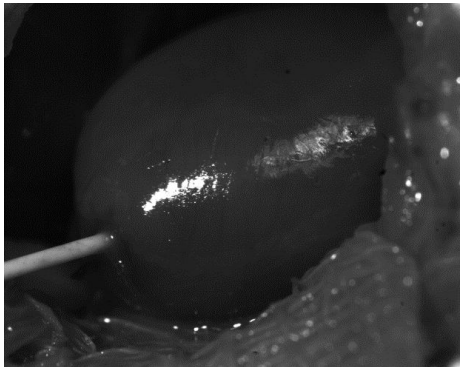
Typical Data Sets for One Experiment – Raw Data

- Images of Heart (mechanical function)
  - 1,000 images/data set
  - ~10 data sets/experiment
- A/D inputs (correspond to images)
- General images
- Histological sections
  - ~3/slide
  - ~500 per experiment



Data Collection

- High frame rate (~250 fps) videos
  - Target: >=30 Frames per cardiac cycle
- 8-bit grayscale
- Speckle application
  - Silicon carbide nanoparticles with retro-reflective beads
  - Increases light intensity distribution



Data Naming Format

- Images
  - Exp#_DS#_frame# .file format
  - Example of such a file is:

    Exp0231_ds2_0008.tif

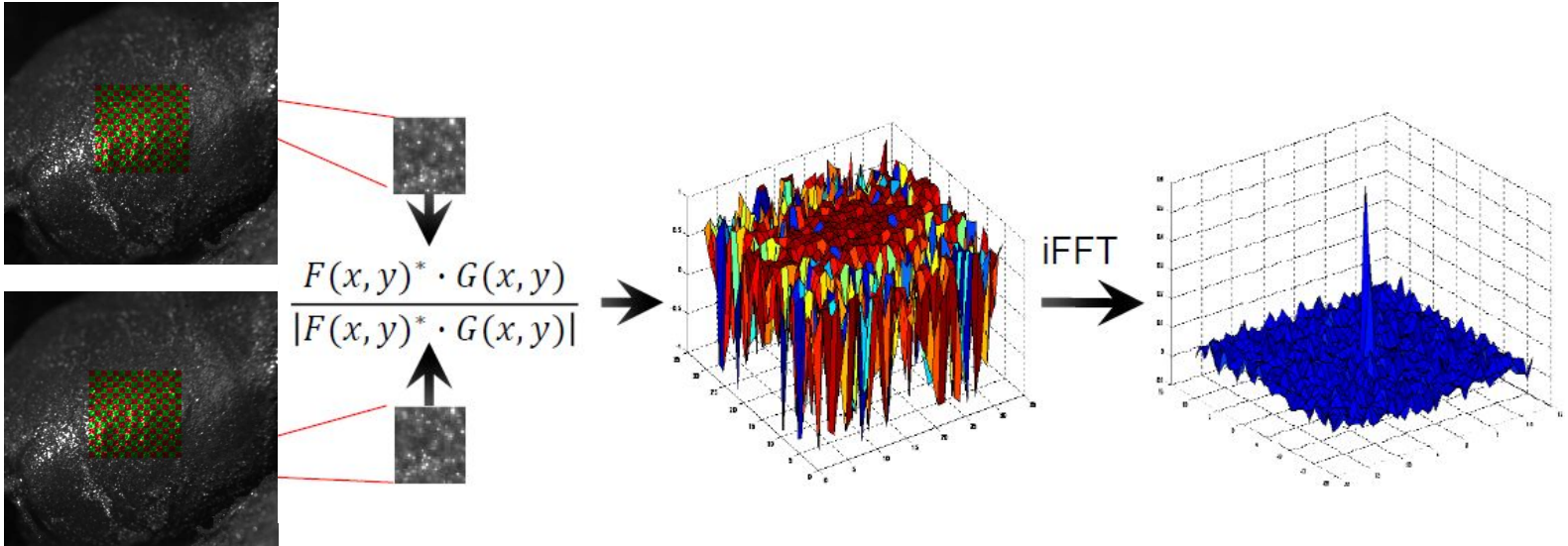
    Experiment #: 0231

    Data Set #: 2

    Frame #: 8
- Pressure data
  - Exp#_DS#.fileformat
  - Example: Exp0231_ds2.pdt

Data Analysis: Mathematics

- Images combined in the Fourier domain using an interference function
- Inverse Fourier transform yields an impulse function
- Displacements can be determined with sub-pixel accuracy

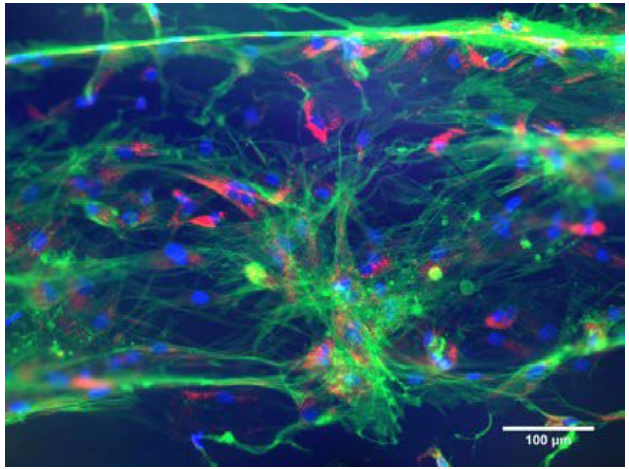


Typical Data Sets for One Experiment – Analyzed Data

- U and V displacement data between 2 images
  - Summary sheet for each heart beat
  - Regional analysis
- Histology
  - H&E; trichrome; phase contrast
  - Immunohistochemistry
- Actinin, troponin, actin, calcium channels, Qdots
- Images
  - 5x, 10x, 40x
  - Panorama
  - Confocal

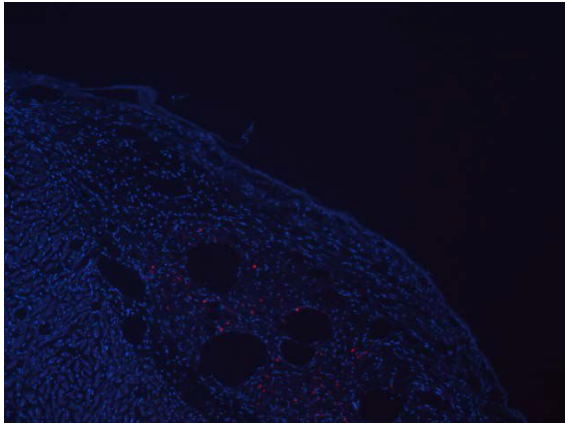
Data Naming Format

Histology

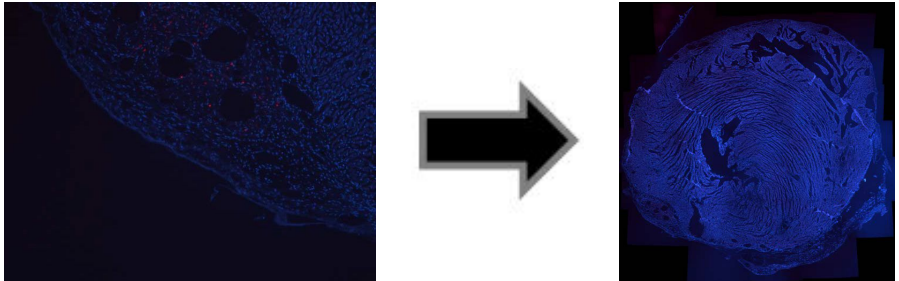


- Apex of the heart -A
- Basal side (base of the heart) -B
- Experiment #, A or B, and Section #
- Example: 0231A_216
- When sample is stained, an abbreviation for the type of stain used is added to the slide
- Example: 0231A_216_act (actinin)

Data Naming Format
Histology



- When doing a panorama
  - Same naming convention
  - Example: 0231A_216_act (actinin)
  - Add:
    - Row – letter (A, B, C, etc.)
    - Column – number (1, 2, 3…)



Students and Data Management

- Review plan with them extensively in beginning of year (2nd time during year)
- Must get their buy in!
- PhD students are critical
  - MS & undergrads report to them
  - They are in lab the longest
- They have little/no training in data management